Chi-Square Analysis

Credit Ratings & Bond Default Outcomes in the Indian Corporate Bond Market

Project Overview

This project applies the Chi-Square Test of Independence to the Indian corporate bond market to determine whether a bond's credit rating is statistically associated with its default outcome. Credit ratings assigned by agencies such as CRISIL and ICRA are intended to communicate default risk; this analysis tests that claim empirically using a contingency table framework.

H_0 (Null Hypothesis)	H_1 (Alternate Hypothesis)
Credit rating and bond default outcome are **independent** — ratings carry no predictive information about default risk.	Credit rating and bond default outcome are **associated** — ratings meaningfully differentiate default risk.

Data & Methodology

Dataset Size	240 bond observations
Design	Balanced — 60 bonds per credit rating category
Rating Agency	CRISIL / ICRA (Indian credit rating agencies)
Table Structure	4 × 5 contingency table (4 ratings × 5 default outcomes)
Test	Chi-Square Test of Independence
Significance Level	$\alpha = 0.05$

Contingency Table — Observed Frequencies (O)

Credit Rating	No Default	Technical Default	Restructured	Partial Default	Full Default	Total
AAA	56	2	1	1	0	60
AA	49	4	3	2	2	60
BBB	38	6	7	5	4	60
BB & Below	20	6	9	8	17	60
Total	**163**	**18**	**20**	**16**	**23**	**240**

Row totals are equal (60 each), reflecting a balanced sampling design across all credit rating categories.

Expected Frequencies (E)

Under H_0, expected frequency for each cell = (Row Total × Column Total) / Grand Total. Since all row totals are equal (60) and the grand total is 240, each expected value = 0.25 × Column Total.

Credit Rating	No Default	Technical Default	Restructure d	Partial Default	Full Default	Total
AAA	40.75	4.5	5.0	4.0	5.75	60
AA	40.75	4.5	5.0	4.0	5.75	60
BBB	40.75	4.5	5.0	4.0	5.75	60
BB & Below	40.75	4.5	5.0	4.0	5.75	60
Total	**163**	**18**	**20**	**16**	**23**	**240**

Chi-Square Computation — $(O - E)^2 / E$

Rating	Outcome	O	E	O – E	(O–E)²	(O–E)²/E
AAA	**No Default**	56	40.75	15.25	232.5625	5.7071
	Tech. Default	2	4.50	-2.50	6.2500	1.3889
	Restructured	1	5.00	-4.00	16.0000	3.2000
	Partial Default	1	4.00	-3.00	9.0000	2.2500
	Full Default	0	5.75	-5.75	33.0625	5.7500
AA	**No Default**	49	40.75	8.25	68.0625	1.6702
	Tech. Default	4	4.50	-0.50	0.2500	0.0556
	Restructured	3	5.00	-2.00	4.0000	0.8000
	Partial Default	2	4.00	-2.00	4.0000	1.0000
	Full Default	2	5.75	-3.75	14.0625	2.4457
BBB	**No Default**	38	40.75	-2.75	7.5625	0.1856
	Tech. Default	6	4.50	1.50	2.2500	0.5000
	Restructured	7	5.00	2.00	4.0000	0.8000
	Partial Default	5	4.00	1.00	1.0000	0.2500
	Full Default	4	5.75	-1.75	3.0625	0.5326
BB & Below	**No Default**	20	40.75	-20.75	430.5625	10.5660
	Tech. Default	6	4.50	1.50	2.2500	0.5000
	Restructured	9	5.00	4.00	16.0000	3.2000
	Partial Default	8	4.00	4.00	16.0000	4.0000
	Full Default	17	5.75	11.25	126.5625	22.0109
					TOTAL	**66.8124**

Statistical Results

Chi-Square Statistic	Degrees of Freedom	Critical Value ($\alpha = 0.05$)	Decision
66.8124	**12**	**21.03**	**Reject H_0**

df = (Rows – 1) × (Columns – 1) = (4 – 1) × (5 – 1) = 3 × 4 = 12 | Since 66.81 > 21.03, we reject H_0 at the 5% significance level.

Key Findings & Interpretation

AAA bonds — Strong outperformance:	56 of 60 bonds recorded no default, against an expected 40.75. Zero full defaults were observed versus the expected 5.75, confirming AAA as the most reliable low-risk category.
BB & Below bonds — Significant distress:	Only 20 bonds avoided default, far below the expected 40.75. Full defaults reached 17 — nearly 3× the expected 5.75. This is the single largest contributor to the chi-square statistic (22.01).
Investment-grade gradient (AA → BBB):	Both categories show moderate deviations, with a clear escalation in default frequency as ratings decline, consistent with a risk-rating spectrum.
Ratings carry informational value:	The test statistically validates that CRISIL/ICRA credit ratings are meaningful predictors of default risk in the Indian corporate bond market.

Conclusion: There is statistically significant evidence ($\chi^2 = 66.81$, df = 12, $\alpha = 0.05$) that credit ratings and bond default outcomes are associated in the Indian corporate bond market. Null hypothesis rejected.

Statistical tool: Chi-Square Test of Independence | Dataset: 240 Indian corporate bonds (simulated for academic demonstration; directional patterns consistent with CRISIL/ICRA rating methodology) | Rating agencies: CRISIL / ICRA | Significance level: $\alpha = 0.05$